Manual

Escalation Messages in Machine Data Collection

MDE-ESK 8.2

Version 1.1.23049

Last changed on: 01.09.2020

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Contents

1 Overview: Escalation Messages in Machine Data Collection 4

2 Available Escalations 5

2.1 Machine status change (MST.MALFUNCTION_OCCURRED) 5

2.2 Cyclic request of the machine status (MST.MALFUNCTION_CONTINUE) 6

2.3 Escalations of the PCC connection at the terminal 6

2.4 Beginning of resource status/malfunction (RES.STATUS_START) 8

2.5 End of resource status/malfunction (RES.STATUS_END)..................................8

2.6 Escalations for cycle deviation (CYCLETIME.*)..9

2.7 Escalation when partitioning changes (MST.PARTITIONING_CHANGED) 12

2.8 Escalations to monitor synchronization of data collection 13

2.8.1 Escalations for asynchronous postings (DLG.ASYNCHRONOUS_MESSAGE and DLG.ASYNCHRONOUS_MESSAGE_END)..14

2.8.2 Escalations for tolerance violations of MDE time stamps (DLG.TS_TOLERANCE_VIOLATION)..15

1 Overview: Escalation Messages in Machine Data Collection

Purpose

HYDRA Escalation Management provides a framework of functions that can be used to forward events that occur or were recorded in HYDRA to individual users or user groups in real time. During the process, Escalation Management takes active steps to ensure users are notified.

After notification, Escalation Management monitors times until acknowledgment by the recipients and until escalation is concluded. Escalations can be forwarded to other users or user groups during processing.

Implementation notes

You use Escalation Management if you would like to have active, real-time notification of specific events in the MDE production environment so that you can react early enough to prevent downtimes and so that efficiency and productivity can be increased.

Integration

The events / escalations already triggered in the MDE environment are reported to central Escalation Management. This forms the framework used to forward the events triggered and to be able to follow up on them.

To notify people, Escalation Management accesses both User administration and the HR master data stored in the system. Notifications can be sent out as e-mails by integrating the local mail server into the system.

Features

- Provision of various escalation messages in the MDE environment, such as machine status changes or machine malfunctions.
- Cyclic monitoring of machine states / statuses that have been forwarded to the escalation framework.
- Event configuration: Configuration of order-related events
- Forwarding of detected events to the escalation framework.

2 Available Escalations

2.1 Machine status change (MST.MALFUNCTION_OCCURRED)

The event is triggered when the status changes to "malfunction" (status with production indicator A = general malfunction, S = malfunction, Z = short-term malfunction, N = malfunction without reason).

Event	IDs	Description
MST.MALFUNCTION_OCCURRED	**MNR.MNR**	Machine
	MST.MST	Machine status
	MNR.MGRP	Machine group
	MNR.KST	Cost center
	MST.BEZK	Machine description
	MSTTXT.MSTTXT	Machine status text
	MST.DAUER	Malfunction duration (in seconds) When the status starts, the duration is always 0, except when the old status was "NOT ASSIGNED".
	MST.OPT:PKENN	Production characteristic: S Malfunction A General disturbance Z Short-term malfunction N Malfunction without reason (in this case, the machine status is 30000) Further production characteristics might exist according to the customer's requirements. Please also see the definitions in the machine status configuration.
	MST.BEM	Comment
	MST.PROGN_DAUER MST.PROGN_BEGIN_DAT MST.PROGN_ZEIB_DAT	Expected (predicted) malfunction time (in seconds) from the dialog as well as the point in time when the predicted downtime was entered.
	MSGPRIO	Priority 1 = highest 2 = high 3 = normal 4 = low 5 =lowest

Event	IDs	Description
	MSGCLASS	Information class/importance I = information W = Warning E = Error
	MSGRCV	Recipient/addressee, e.g. group of plant managers

This escalation supports the additional placeholders MSGPRIO, MSGCLASS and MSGRCV.
They are separately described in the documentation entitled HYD-ESK.

2.2 Cyclic request of the machine status (MST.MALFUNCTION_CONTINUE)

The event is triggered cyclically for all machines that are currently assigned to a malfunction status according to the production indicator (A = general malfunction, S = malfunction, Z = short-term malfunction, N = malfunction without reason)

By defining the condition in the escalation management module, a notification may occur, for example, if a particular status exists for a longer time than specified.

Event	IDs	Description
MST.MALFUNCTION_CONTINUE	**MNR.MNR**	Machine
	MST.MST	Machine status
	MNR.MGRP	Machine group
	MNR.KST	Cost center
	MST.BEZK	Machine description
	MSTTXT.MSTTXT	Machine status text
	MST.DAUER	Malfunction duration (in seconds)
	MST.OPT:PKENN	Production characteristic

The MST.DAUER field shows the malfunction time in seconds. When the escalation is configured, this value can be used to specify after which malfunction time the escalation is to be sent at all.

2.3 Escalations of the PCC connection at the terminal

Please note: Only MW 2.0 supports the PCC connection at the terminal.

→→All required channels (MDE, PDV, DNC ...) are initialized when the terminal is being restarted. The escalation ERRPRO.ERROR_PROTOCOL_WRITTEN (see the HYD-ESK documentation) is sent to the server if the assignment for a channel is missing within the PCC connection. In addition, an escalation is also sent if a channel notifies an error while operation is running. The PCC error codes (see the HYD-RET document, error codes as of 4000) are entered in the ERRPRO.ERRCODE field. "PCC" is always entered in the field ERRPRO.EREIG and "TNR" in the field ERRPRO.ERRCLASS. The ERRPRO.BEM field shows the description of the error including the channel number. If available, the connected machine number is entered in the ERRPRO.MNR field, the operation number is optionally entered in the ERRPRO.ANR field.

The following additional steps are performed when the terminal triggers an escalation:

- The escalation is recorded locally in the prot_esc.txt file at the terminal
- The escalation is displayed in a popup dialog at the terminal
- The escalation is sent to the server

To limit the number of terminal escalations, a minimum period of time can be specified for every channel, which has to pass before the next escalation is sent to the server. This minimum waiting time is specified, in seconds, in the file ctwin.ini in the [DLL] section:

EscalationSendBlockInterval=180

If the option “EscalationViewBlock=OFF” is also set in the same section, the display of the popup window is also suppressed for the specified interval at the terminal.

It is also possible to completely deactivate the popup dialog and messages to the server:

EscalationPopup=OFF

→ Do not show notification at the terminal if an error occurs

EscalationSend=OFF

→ Do not send messages to HYDRA that trigger an escalation

2.4 Beginning of resource status/malfunction (RES.STATUS_START)

The event is triggered if a parallel (resource) status is set (RES_STB dialog).

Event	IDs	Description
RES.STATUS_START	RES.RES	Resource
	RES.TYP	Resource type
	STA.STA	Status
	STA.TYP	Status type of the status
	STA.DAUER	Fix 0
	STA.TXT	Status text
	MSGPRIO	Priority 1 = highest 2 = high 3 = normal 4 = low 5 =lowest
	MSGCLASS	Information class/importance I = information W = Warning E = Error
	MSGRCV	Recipient/addressee, e.g. group of plant managers

Please note

This escalation supports the additional placeholders MSGPRIO, MSGCLASS and MSGRCV. These fields are only sent to the escalation, provided that data is included within dialog data. They are separately described in the documentation entitled HYD-ESK.

2.5 End of resource status/malfunction (RES.STATUS_END)

The event is triggered if a parallel (resource) status is completed (RES_STB / RES_STE dialog).

Event	IDs	Description
RES.STATUS_END	RES.RES	Resource
	RES.TYP	Resource type
	STA.STA	Status
	STA.TYP	Status type of the status
	STA.DAUER	Duration with RES.STATUS_END

Event	IDs	Description
	STA.TXT	Status text
	MSGPRIO	Priority 1 = highest 2 = high 3 = normal 4 = low 5 =lowest
	MSGCLASS	Information class/importance I = information W = Warning E = Error
	MSGRCV	Recipient/addressee, e.g. group of plant managers

Please note

This escalation supports the additional placeholders MSGPRIO, MSGCLASS and MSGRCV. These fields are only sent to the escalation, provided that data is included within dialog data. They are separately described in the documentation entitled HYD-ESK.

2.6 Escalations for cycle deviation (CYCLETIME.*)

If the recorded actual cycle deviates from the target cycle of the machine, an escalation management event can be triggered and an employee or a defined group of people will be informed about the cycle deviation.

The cycle parameters at the machine (MDE menu: master data à machines à cycle parameters) define whether or not an escalation is to be generated. The action limits and tolerance limits are defined for cycle recording. An escalation can be triggered respectively if these limits are exceeded.

The following events may be triggered:

Event (in escalation management)	Action triggering the event
CYCLETIME.POS_TOLERANCE_LIMIT_EXCEEDED	Positive tolerance limit has been exceeded
CYCLETIME.NEG_TOLERANCE_LIMIT_EXCEEDED	Negative tolerance limit has been exceeded
CYCLETIME.POS_ACTION_LIMIT_EXCEEDED	Positive action limit has been exceeded
CYCLETIME.NEG_ACTION_LIMIT_EXCEEDED	Negative action limit has been exceeded

Details on the function

The configured process parameters of the machine are the basis for specifying an escalation. These parameters are the target cycle of the machine, which is normally specified by the logged on OP(s) as well as the collected actual cycle, which is usually determined by the shop floor terminal.

The actual cycle is transferred cyclically from the terminal to the server (for further details on this please refer to the documentation entitled “processing and configurations in HYDRA-MDE 7.2”, section “determination of the actual cycle”). While posting the actual cycle, it is also checked whether or not an escalation is to be triggered.

In general, an escalation is only triggered if

- cycle parameters are configured for the machine
- the (current) target cycle of the machine is > 0
- the machine is in the “production” status, i.e. a status that posts on RPA 11 (main/principal utilization time) is available at the machine

The minimum cycle (configuration setting at the machine) does not affect triggering of escalations.

An escalation is only triggered if the limits are exceeded/not reached. Escalations are not triggered when the limits are reached.

Escalations are only triggered if the limit is exceeded/not reached for the first time. In case an actual cycle violating the same limit is recorded subsequently, this (same) escalation is not triggered anymore.

In case a collected actual cycle exceeds/does not reach both limits (action limit and tolerance limit) in one collection process, only the escalation referring to the tolerance limit is triggered.

But both escalations are triggered if the recorded actual cycle successively exceeds/does not reach both limits (the action limit at first and then the tolerance limit).

If the collected actual cycle exceeds/does not reach the tolerance limit and then drops/increases successively until reaching the action limit, the escalation for the action limit will not be triggered.

Whether or not an OP is active, generally does not affect the triggering of an escalation. Provided that the user wishes to trigger escalations only if an OP is active, the target cycle of the machine has to be set to 0 when the last OP of the machine is interrupted/logged off. If required, this function can be made available during customizing of HYDRA.

Furthermore, the event(s) has/have to be configured correctly within escalation management. For further details on this, please refer to the document dealing with the HYDRA Escalation Management.

The following parameters are available for all four events:

Event	**IDs**	**Description**
CYCLETIME.POS_TOLERANCE_LIMIT_EXCEEDED or CYCLETIME.NEG_TOLERANCE_LIMIT_EXCEEDED or CYCLETIME.POS_ACTION_LIMIT_EXCEEDED or CYCLETIME.NEG_ACTION_LIMIT_EXCEEDED	**MNR.MNR**	Machine
	MNR.MGRP	Machine group
	MNR.KST	Cost center of the machine
	MST.BEZK	Short name
	MST.BEZL	Designation (comment)
	MNR.FIR	Company
	MNR.VAB	Responsibility area
	ZYP.LIMIT	Cycle limit in percent that has been exceeded
	ZYP.DEVIATION	Current deviation between actual cycle and target cycle (in percent)
	MNR.SZY	Target cycle in sec/1000 including decimal places
	MNR.SZY_ASTROKE	Target cycle in sec/1 including decimal places
	MNR.IZY	Actual cycle in sec/1000 including decimal places
	MNR.IZY_ASTROKE	Actual cycle in sec/1 including decimal places

Please note: The machine number is the key of the escalation

<u>Examples for triggering escalations on the basis of the upper action and tolerance limit:</u>

Target cycle	**Cycle parameter**		**Computed limit values relating to the target cycle**		**Actual cycle**	**Escalation**	
	Upper action limit	Upper tolerance limit	Upper action limit	Upper tolerance limit		Upper action limit	Upper tolerance limit
22560	5	10	23688	24816	22500		
22560	5	10	23688	24816	22600		
22560	5	10	23688	24816	23688		
20000	5	10	21000	22000	22500		✓
20000	5	10	21000	22000	20400		
20000	5	10	21000	22000	21005	✓	

Target cycle	Cycle parameter		Computed limit values relating to the target cycle		Actual cycle	Escalation	
	Upper action limit	Upper tolerance limit	Upper action limit	Upper tolerance limit		Upper action limit	Upper tolerance limit
20000	5	10	21000	22000	22400		✓
20000	5	10	21000	22000	22280		
20000	5	10	21000	22000	22210		
20000	5	10	21000	22000	21550		
20000	5	10	21000	22000	20120		
20000	5	10	21000	22000	21550	✓	
20000	5	10	21000	22000	21550		
20000	5	10	21000	22000	22400		✓

Please note:

The values are indicated in seconds per 1000 cycles.

green Actual cycle within normal range

yellow Actual cycle outside of the action limit (but within the tolerance limit)

red Actual cycle outside of the tolerance limit

2.7 Escalation when partitioning changes (MST.PARTITIONING_CHANGED)

In general, an escalation is only triggered if

- portioning is changed manually (using the M_TLG dialog)
- partitioning of the machine has actually changed

The escalation is triggered every time partitioning is changed manually, even if there is still another open escalation on this subject for this machine!

Furthermore, the event(s) has/have to be configured correctly within escalation management. For further details on this, please refer to the document dealing with the HYDRA Escalation Management.

The below-mentioned escalation parameters are available:

Event	IDs	Description
MST.PARTITIONING_CHANGED	MNR.MNR	Machine
	MNR.MGRP	Machine group
	MNR.KST	Cost center of the machine
	MST.BEZK	Short name
	MST.CAT	Category
	MNR.ART	Type
	MNR.VAB	Responsibility area
	MNR.TLG:OLD	Old machine partitioning
	MNR.TLG:NEW	New machine partitioning
	ANR.ANR	Order
	ANR.RES:WNR	Main tool of the order
	ANR.PNR	Personnel number
	ANR.KNR	Badge number
	ANR.TLG:OLD	Old partitioning of the order
	ANR.TLG:NEW	New partitioning of the order

2.8 Escalations to monitor synchronization of data collection

Multichannel data acquisition (i.e. postings deriving from different channels for a machine), such as MDE data deriving from a PCC and order postings deriving from an AIP or MOC, might lead to synchronization problems if one of the two input components is no longer synchronous. Consequently, quantities and statuses are posted incorrectly for orders.

The below-mentioned escalations indicate that such synchronization problems have occurred (and when). Consequently, data can be corrected manually and specifically (e.g. in the order or machine-related HYDRA postings).

These escalations require customizing:

- The function is available as of hymw.exe/out V8.1.1.484.
- The below-mentioned INI configurations are created initially along with the initial data for escalations by the patch *db_sql\dbp_mde_esk_synchronicity.hsc* and are inactive. The DB patch has to be called explicitly.

HYDRA needs to be restarted if the HYDRA INI configuration is changed, in order for these changes to be taken over by the interface processes.

2.8.1 Escalations for asynchronous postings (DLG.ASYNCHRONOUS_MESSAGE and DLG.ASYNCHRONOUS_MESSAGE_END)

The escalations *DLG.ASYNCHRONOUS_MESSAGE* and *DLG.ASYNCHRONOUS_MESSAGE_END* ("delayed MDE postings", "MDE lags behind or is ahead of time") are initiated, once the first posting is asynchronous or if the next posting is again synchronous. A posting is considered being asynchronous if its time stamp deviates from the current server time by more than a defined period of time "max. time deviation". Only posting dialogs are checked (even if they are recorded OFFLINE). Checking does not take place when switching from/to daylight saving time (i.e. between 2.00 a.m. and 3.00 a.m. on the two concerned days in autumn and spring).

The "max. time deviation" is defined in the INI configuration:

INI	MDE	
SECTION	ASYNCHRONOUS_MESSAGE	
IDENT	TIME_INTERVAL	
VALUE	<time in seconds; greater than 180>	If values less than 180 seconds are configured, the 180 seconds deviation will be checked
AKTIV	J	J: enabled N: disabled

Furthermore, the event(s) has/have to be configured correctly within Escalation Management.

The following escalation parameters are available:

Event	ID	Description
DLG.ASYNCHRONOUS_MESSAGE *or* DLG.ASYNCHRONOUS_MESSAGE_END	DLG.DLG	Dialog
	DLG.DAT	Dialog date
	DLG-ZEI	Dialog time
	DLG.USR	User
	MNR.MNR	Machine
	ANR.ANR	Operation (opt.)
	PNR.PNR	Personnel number (opt.)
	PNR.KNR	Badge number (opt.)

2.8.2 Escalations for tolerance violations of MDE time stamps (DLG.TS_TOLERANCE_VIOLATION)

The escalation *DLG.TS_TOLERANCE_VIOLATION* is initiated if an order posting that has been checked for validity is entered and exceeds the "max. tolerance duration" (difference between MDE time stamp of the machine and the time stamp of dialog data).

Only the below-mentioned posting dialogs are checked (even if they are recorded OFFLINE). PCCMD postings are not checked.

- Log OP on (A_AN)
- Log OP and person on (A_P_AN)
- Partial upload (A_TR)
- Interrupt OP (A_UN)
- Log OP off (A_AB)
- Quantity upload (A_MR)
- Finish OP (A_BE)
- Log merged OP on (SA_AN)
- Partial upload of merged OP (SA_TR)
- Interrupt merged OP (SA_UN)

- Log merged OP off (SA_AB or S_ABME)
- Log person on (P_AN)
- Log person off (P_AB)
- Log all persons off (P_AAB)

Checking does not take place when switching from/to daylight saving time (i.e. between 2.00 a.m. and 3.00 a.m. on the two concerned days in autumn and spring).

The machine’s MDE time stamp is checked, which is only recorded on the PCC (terminal type PCCMD).

The “max. tolerance duration” is defined in the INI configuration. The time stamp is not checked and no escalation is sent if an active INI configuration does not exist for the TOLERANCE_PERIOD.

INI	MDE	
SECTION	TS_TOLERANCE_VIOLATION	
IDENT	TOLERANCE_PERIOD	
VALUE	<Time in seconds; greater than 180>	If values less than 180 seconds are configured, the 180 seconds deviation will be checked
AKTIV	J	J: enabled N: disabled

The ”reaction“ is defined in the INI configuration. If the INI configuration is not active for TOLERANCE_REACTION, “W” will be assumed to be the default value. The TOLERANCE_REACTION will only be evaluated if the TOLERANCE_PERIOD is configured.

INI	MDE	
SECTION	TS_TOLERANCE_VIOLATION	
IDENT	TOLERANCE_REACTION	
VALUE	E/W/N	E: posting is to be rejected W: only a warning is sent N: no reaction
AKTIV	J	J: enabled N: disabled

If one of the options “E“ or “W“ is to be used when tolerances are not respected, the terminal has to be configured in such way as for it to no longer perform offline postings. This can be realized by the option “checking required” in the terminal label (BAPI ID TNR.PLAUS:OFF=J).

If “W” is configured as “reaction”, the terminal shows the message “The point in time of the posting differs too much from the machine's posting status!”. The user has to confirm this message manually. MOC does not show a message.

If “E” is configured as “reaction”, the posting is rejected by the error code 95 (short text: “Posting beyond sync.!; long text: “Posting beyond synchronization!”).

The period of time for displaying the warning message (for reaction = “W”) is set to 0 by default, i.e. in this case the user has to confirm the posting manually. As an alternative, a time may be indicated after that the message will be closed automatically.

INI	MDE	
SECTION	TS_TOLERANCE_VIOLATION	
IDENT	DISPLAY_TIME_WARNING	
VALUE	{time in seconds}	0: The user has to confirm the message > 0: The message is shown for the maximum duration in seconds.
AKTIV	J	J: enabled N: disabled

Furthermore, the event(s) has/have to be configured correctly within Escalation Management.

The following escalation parameters are available:

Event	**ID**	**Description**
DLG.TS_TOLERANCE_VIOLATION	DLG.DLG	Dialog
	DLG.DAT	Dialog date
	DLG.ZEI	Dialog time
	MNR.DAT	Status date
	MNR.ZEI	Status time
	DLG.USR	User
	MNR.MNR	Machine
	ANR.ANR	Operation (opt.)
	PNR.PNR	Personnel number (opt.)
	PNR.KNR	Badge number (opt.)